TEMASEK JUNIOR COLLEGE
Preliminary Examinations 2009

BIOLOGY **8875/01**
Higher 1 **Friday, 2 October**
Paper 1 Multiple Choice **1 hour**

Additional Materials: **Multiple Choice Answer Sheet**

READ THESE INSTRUCTIONS FIRST

Write in a soft pencil
Do not use staples, paper clips, highlighters, glue or correction fluid.
Write your name, Centre number and index number on the Answer Sheet in the spaces provided.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.
Choose the **one** you consider correct and record your choice in soft pencil on the separate Answer Sheet

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for the wrong answer.
Any rough working should be done in this booklet.

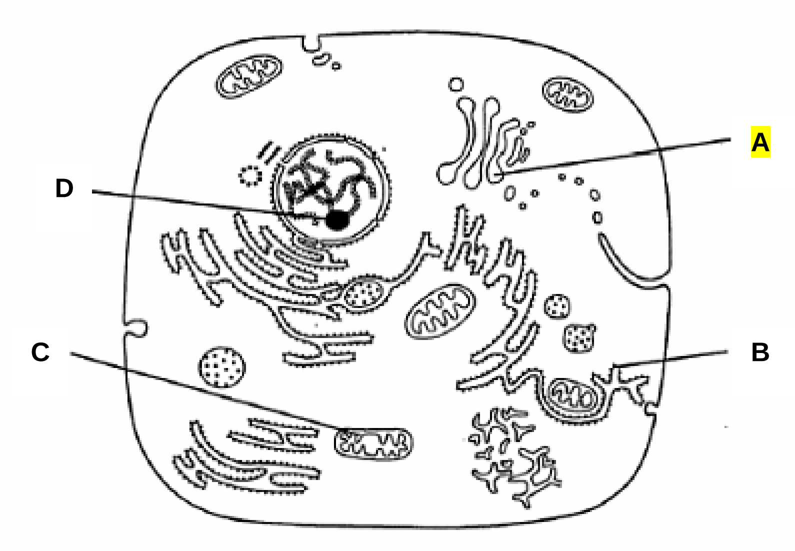
This document consists of **14** printed pages.

1. Which group of structures would be clearly seen in a suitably stained plant cell under the light microscope?

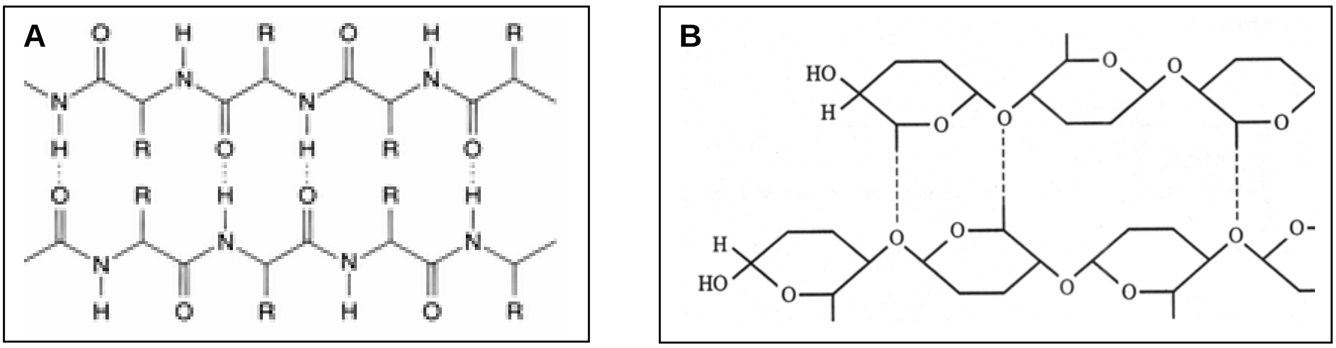
A. Chloroplasts, mitochondria, nuclei and ribosomes
B. Chloroplasts, nuclei, starch grains and vacuole
C. Endoplasmic reticulum, Golgi apparatus, mitochondria and nuclei
D. Endoplasmic reticulum, ribosomes, starch grains and vacuole

2. The diagram below shows the ultrastructure of a eukaryotic cell.

Which organelle does not contain nucleic acid?



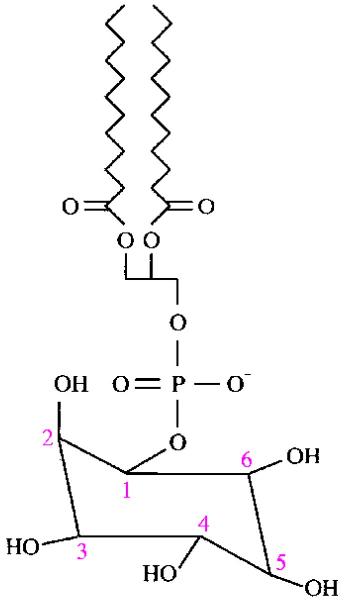
3. The following shows two molecules that perform structural functions.



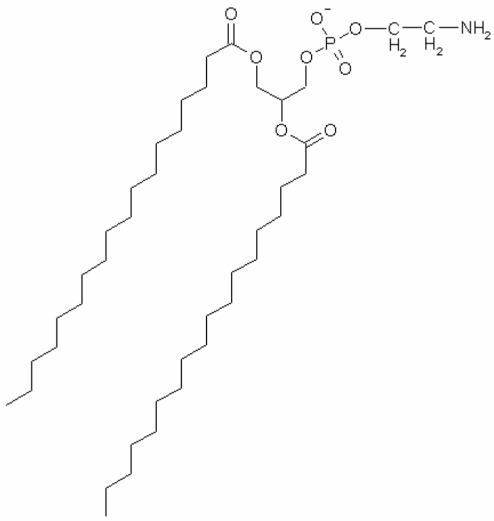
Which of the following statements does **not** describe a similarity between molecules **A** and **B**?

A. Hydrogen bonds form cross-linkages between neighboring chains
B. Monomers are joined together via peptide bonds
C. Monomers can be extracted by hydrolysing the bonds within each chain.
D. Disulphide bonds are present in both molecules.

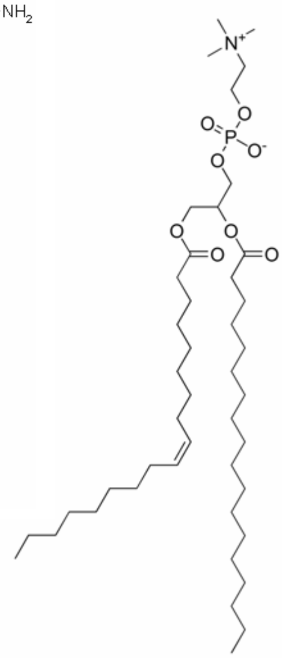
4. The diagram below shows 3 different phospholipids, phosphatidylcholine, phosphatidylethanolamine and phosphatidylinositol.



phosphatidylinositol



phosphatidylethanolamine



phosphatidylcholine

Which of the following statements is correct?

A. Membranes with a higher phosphatidylcholine content are more fluid
B. Phosphatyidylinositol is unable to form a bilayer in an aqueous environment.
C. Only phosphatyidylethanolamine shows ambivalent behaviour towards water.
D. Only phosphatidylinositol has 2 fatty acid side chains.

5. Which of the following options correctly matches the functional and structural features of cellulose, collagen, glycogen and triglycerides?

		Function	Structure		
			Fibrous	Molecule held together by hydrogen bonds	Branched chains
A.	Cellulose	Support	✓	✗	✓
	Collagen	Strengthening	✓	✓	✗
B.	Cellulose	Support	✓	✓	✗
	Triglyceride	Energy source	✗	✗	✗
C.	Collagen	Strengthening	✓	✓	✓
	Glycogen	Storage	✗	✗	✓
D.	Glycogen	Storage	✗	✓	✓
	Triglyceride	Energy source	✗	✓	✗

6. The table shows some differences between glucose and collagen.

Which difference explains the fact that β-glucose is soluble in water, while collagen is insoluble in water?

Difference	β-glucose	collagen
A.	does not contain hydrogen bonds	contain hydrogen bonds
B.	monomer	polymer
C.	absence of R-groups	presence of R-groups
D.	contains polar group	contains outward-facing hydrophobic R-groups

7. Which combination of bonds contributes to the structure of the following molecules?

	Amylase	**Amylopectin**	**Glucagon**	**Glycogen**
A.	ionic bond	α (1,6) glycosidic bond	peptide bond	α (1,4) glycosidic bond
B.	phosphodiester bond	α (1,4) glycosidic bond	hydrogen bond	α (1,6) glycosidic bond
C.	α (1,4) glycosidic bond	α (1,6) glycosidic bond	α (1,6) glycosidic bond	β (1,4) glycosidic bond
D.	covalent bond	β (1,4) glycosidic bond	hydrophobic interactions	α (1,6) glycosidic bond

8. Beetroot cells contain a water-soluble red pigment. Two test-tubes were set up as described in the table.

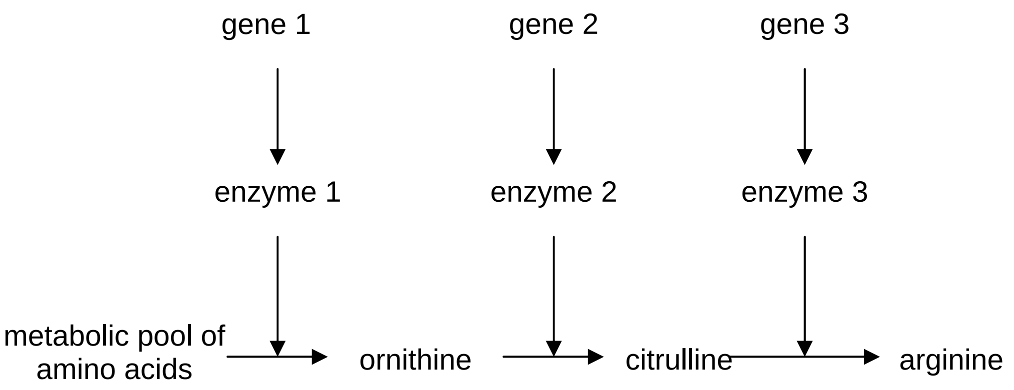
tube 1	pieces of washed raw beetroot in water
tube 2	pieces of washed raw beetroot in water containing 3 drops of respiratory inhibitor

After 30 minutes, the water in tube 2 contained red pigment but the water in tube 1 did not.

Which statement explains the colour of the water in tube 2?

A. Pigment molecules passed out and were replaced by the inhibitor molecule.
B. The cell membranes were unable to retain the pigment.
C. The cell walls were made permeable to the pigment
D. Water passed out of the cells by osmosis and carried the soluble pigment with it.

9. The diagram shows a metabolic pathway involving some amino acids.



Three mutant strains of *Escherichia coli* each fail to produce a different enzyme in the pathway.

What should be supplied to the culture medium to enable growth of all three mutant strains?

A. arginine
B. citrulline
C. citrulline and arginine
D. citrulline and ornithine

10. The four acids shown below form part of an enzyme-catalysed sequence.

oxoglutaric acid (α-keto glutaric acid) $\xrightarrow{\text{enzyme 1}}$ succinic acid $\xrightarrow{\text{enzyme 2}}$ fumaric acid $\xrightarrow{\text{enzyme 3}}$ malic acid

The addition of malonic acid results in no change in the concentration of oxoglutaric acid, an accumulation of succinic acid, and a near absence of both fumaric acid and malic acid.

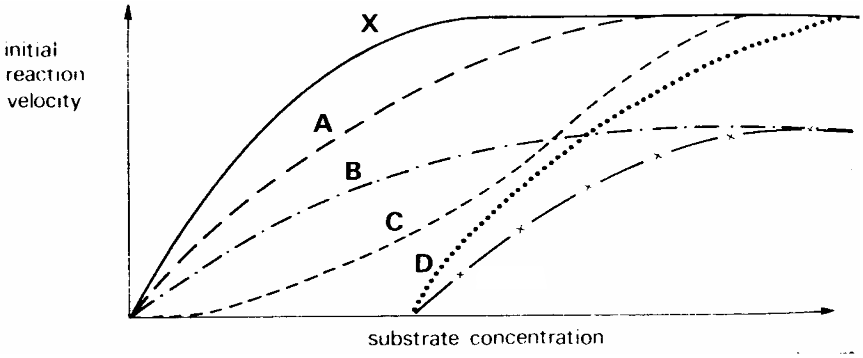
Further addition of fumaric acid results in the formation of malic acid.

What does this information indicate about malonic acid?

A. It is an inhibitor of enzyme 1.
B. It is an inhibitor of enzyme 2.
C. It catalyses the formation of succinic acid.
D. It inhibits enzyme 3.

[Turn over

11. In the graph below, **X** represents the relationship between an enzyme and the concentration of its substrate under optimal conditions and without an inhibitor.



Which of the curves, **A**, **B**, **C** or **D** represents the expected result when the same experiment is conducted in the presence of a fixed, low concentration of non-competitive inhibitor?

12. The events below occur during different phases of mitosis.

1 The chromatids become visible.
2 The centromeres align on the 'equator' of the spindle.
3 The nuclear membranes form.
4 DNA replicates.
5 The sister chromatids separate.

Which one of the following shows the correct sequence of evens occurring during each phase?

	interphase	*prophase*	*metaphase*	*anaphase*	*telophase*
A	1	2	5	4	3
B	1	4	2	3	5
C	3	4	1	2	5
D	4	1	2	5	3

13. Which one of the following structures would be found in an animal cell undergoing mitosis, but not in a plant cell undergoing mitosis?

A. centriole
B. centromere
C. chromatid
D. telomere

14. Adenine comprises 0.32 of the nitrogenous bases in the DNA of cells from a bacterial clone.

What is the proportion of guanine in the DNA?

A. 0.12
B. 0.18
C. 0.32
D. 0.36

15. An experiment is carried out to show that DNA replication is semi-conservative.

Escherichia coli cells whose DNA contains only 'heavy' nitrogen (^{15}N) are grown in a culture medium containing the normal isotope of nitrogen (^{14}N) and allowed to divide three times.

What will be the percentage of DNA molecules containing both forms of nitrogen after the third division?

A. 12.5%
B. 25%
C. 50%
D. 100%

16. Which of the functions of RNA is incorrectly matched?

A. Transfer RNAs bind with mRNA to facilitate translation.
B. Messenger RNAs encode information for synthesis of polypeptides.
C. Small nuclear RNAs bind with ribonucleoproteins to form spliceosomes.
D. Ribosomal RNAs bind with transfer RNA to catalyse the formation of phosphodiester bonds.

17. Carbon dioxide labelled with ^{14}C has been used to identify the intermediate compounds in the Calvin cycle, the light-independent stage in photosynthesis.

Which compound would be the first to contain the ^{14}C?

A. triose phosphate
B. GP (PGA)
C. RuBP
D. glucose

[Turn over

18. The diagram shows part of the normal sequence of an mRNA molecule.

CCAUAGUGGUCCGUAAAAUGGC

A mutation in the DNA resulted in a polypeptide beginning with the following sequence.

Glycine – serine – proline – glycine - isoleucine

The DNA triplets for some amino acids are:

glycine	proline	leucine	serine	isoleucine
CGA GGT GGC	CCA CCG	TTA CTT CTC	TCA TCG	ATA ATT

Which mutation has occurred in the DNA molecule?

A. the loss of a nucleotide
B. an addition of an extra nucleotide
C. a reversal in the order of nucleotides
D. the replacement of one nucleotide by a different nucleotide

19. Which of the following is **incorrectly** matched with its function?

A. Promoter – where RNA polymerase first binds to DNA
B. Regulatory gene – binds to the repressor protein
C. Structural gene – synthesizes mRNA by transcription
D. Operator – if unbound, allows RNA polymerase to bind to DNA

20. The diagram below shows part of a gene in a human cell.

Exon	Intron	exon

In one form of the blood condition thalassaemia, there is a mutation in an intron of the haemoglobin gene.

Which stage of gene expression is most likely to be directly affected by this mutation?

A. Splicing of pre-mRNA.
B. Attachment of mRNA to a ribosome.
C. Binding of RNA polymerase to DNA.
D. Activation of the gene by a steroid hormone.

21. A strain of toad has only one nucleolus in the nucleus of each cell instead of the usual two. When toads with one nucleolus per cell are mated, approximately a quarter of the offspring have two nucleoli per nucleus, half have one nucleolus per nucleus and a quarter have no nucleoli.

What is the most likely explanation of these results?

A. The possession of one nucleolus is due to autosomal linkage.
B. The possession of one nucleolus is due to the heterozygous condition.
C. The allele for the presence of two nucleoli is recessive.
D. The allele for the presence of two nucleoli is dominant.

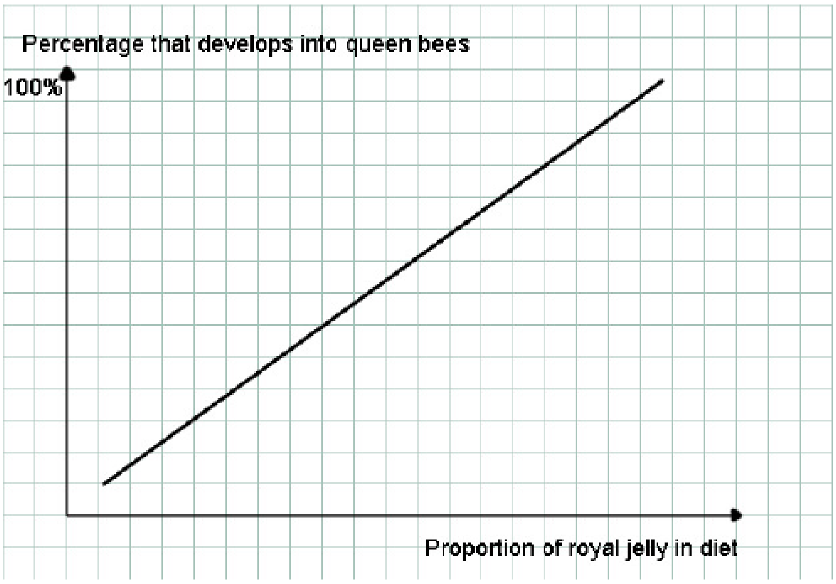
22. The allele for an enzyme involved in the production of chlorophyll in geraniums is C^g. The mutant allele C^w codes for a defective gene with no pigment. A cross between two heterozygotes produced pale green plants and dark green plants in the ratio 2:1.

What is the most likely explanation of this ratio?

A. C^g is a dominant allele.
B. C^w is a dominant allele.
C. $C^w C^g$ is a lethal phenotype.
D. $C^w C^w$ is a lethal phenotype.

[Turn over

23.



Which of the following statement best explains the graph?

A. Worker bees develop into queen bees over time.
B. The phenotype of the honeybee is dependent on diet.
C. Worker bees develop into queen bees only when fed exclusively on royal jelly.
D. Having a higher proportion of royal jelly in the diet causes genetic changes in honeybees.

24. Six tubes were set up as shown in the table.

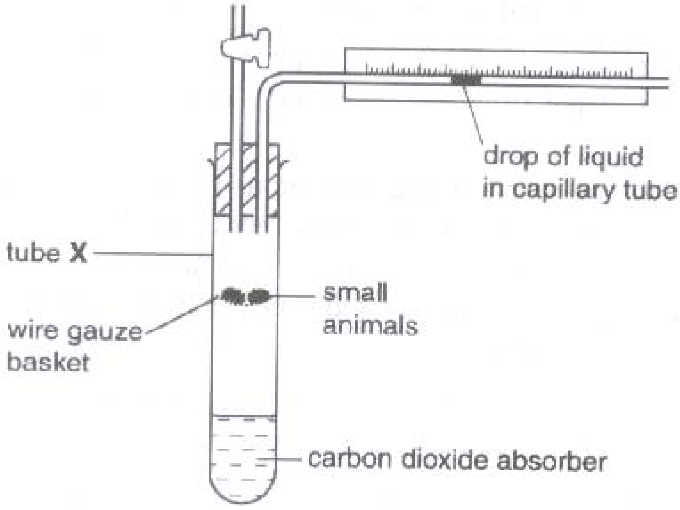
tube	contents
1	Glucose + homogenized plant cells
2	Glucose + mitochondria
3	Glucose + cytoplasm lacking organelles
4	Pyruvate + homogenized animal cells
5	Pyruvate + mitochondria
6	Pyruvate + cytoplasm lacking organelles

After incubation, each sample was analysed to determine the presence of carbon dioxide and ethanol.

In which tube(s) is lactate most likely to be present?

A. 1 and 3 only
B. 2, 3, 5 and 6 only
C. 4, 5, and 6 only
D. 3 and 6 only

25. The diagram shows a simple respirometer.



The changes in gas volume in the tube are measured at intervals.

time (minutes)	gas volume with carbon dioxide absorber (cm^3)	gas volume without carbon dioxide absorber (cm^3)
0	0.0	0.0
10	-0.4	-0.1
20	-0.8	-0.2
30	-1.2	-0.3

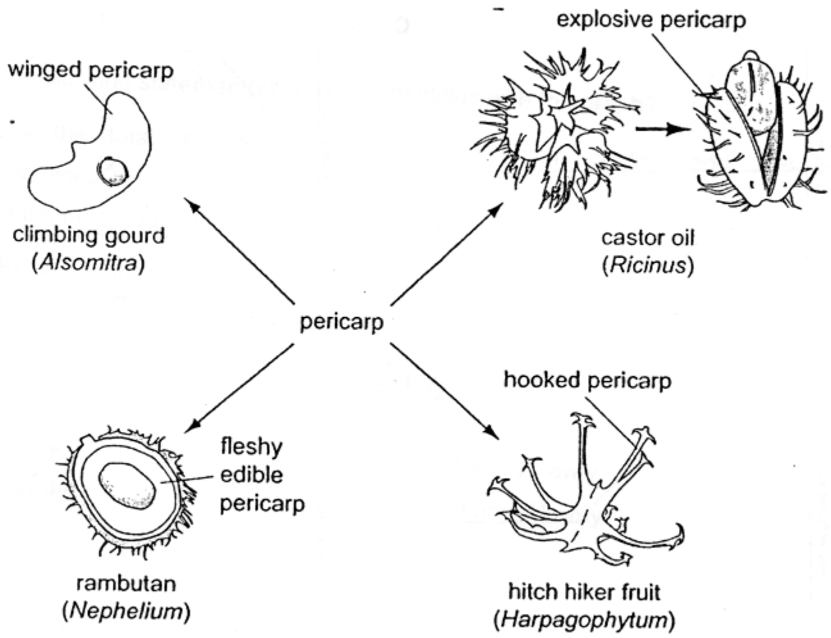
Tube **X** contains 2 g of small animals. What is the carbon dioxide output per g per hour for these organisms?

A. 0.9 cm^3
B. 1.8 cm^3
C. 2.4 cm^3
D. 4.8 cm^3

26. Which of the following has evolved mainly as a result of artificial selection?

A. Darker colouring of the peppered moth near individual areas
B. Increased production of antibiotics by the fungus *Penicillium* species
C. Increased resistance of houseflies to the insecticide DDT
D. Increased tolerance of lichens to heavy metals on the bark around mine workings

27. The diagram illustrates variation in the pericarp (fruit wall) for a variety of methods of seed dispersal.



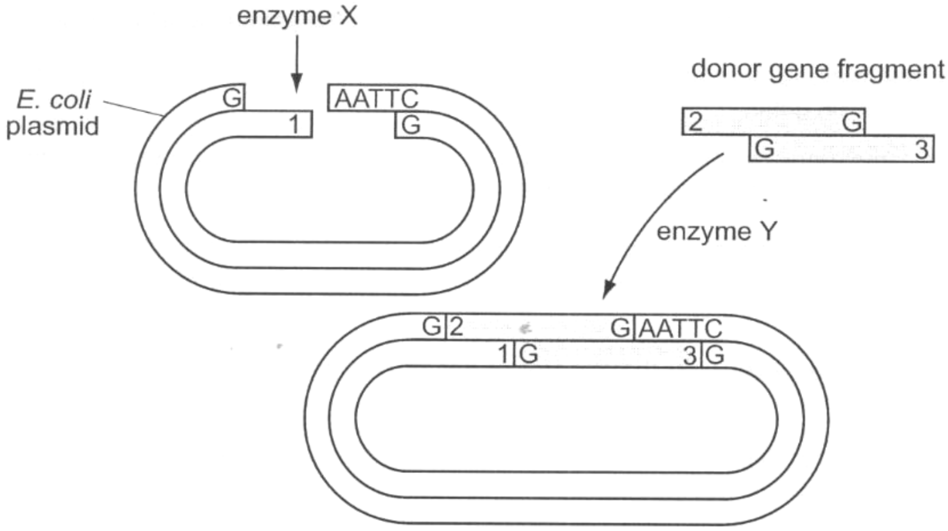
What do these examples illustrate?

A. The adaptive radiation of analogous structures showing divergent evolution.
B. The adaptive radiation of analogous structures showing convergent evolution.
C. The adaptive radiation of homologous structures showing divergent evolution.
D. The adaptive radiation of homologous structures showing convergent evolution.

28. What are the normal functions of stem cells in a living human?

	Functions		
A.	Differentiating into many kinds of cells in a 3 – 5 day old human embryo	Producing insulin in pancreases damaged by type 1 diabetes	Cells that can differentiate into bone cells in the skeletons
B.	Differentiating into many kinds of cells in a 3 – 5 day old human embryo	Producing red blood cells worn out by normal wear and tear	Cells that can differentiate into cardiac muscle cells in the heart
C.	Differentiating into only one kind of cells in a 3 – 5 day old human embryo	Producing dopamine in the brains of people with Parkinson’s	Cells that can differentiate into cartilage cells in the joints
D.	Differentiating into only one kind of cells in a 3 – 5 day old human embryo	Producing differentiated cells that can be used to screen new drugs	Cells that can differentiate into nerve cells in the brain

29. The diagram shows some stages in making recombinant DNA using *E.coli* plasmids.

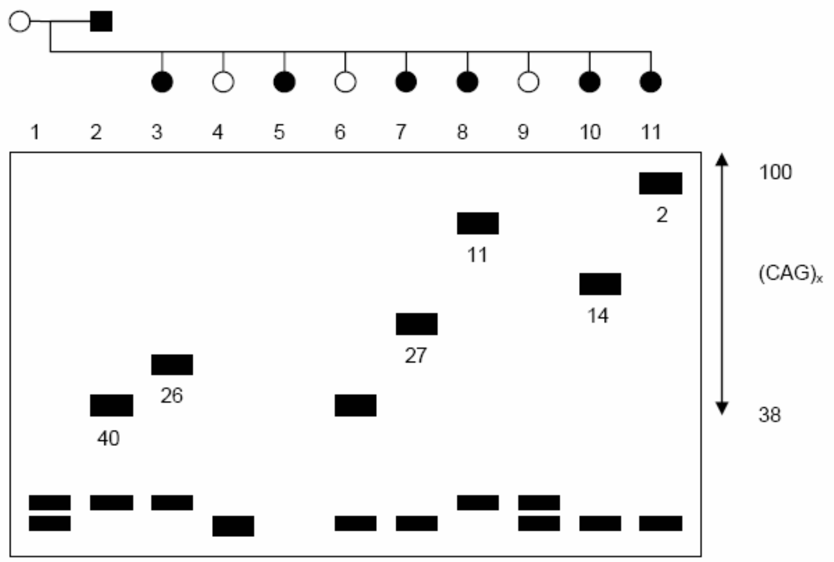


Which are the correct enzymes X and Y and order of the bases at 1, 2 and 3?

	X	Y	1	2	3
A.	Ligase	Restriction enzyme	TTAA	AATT	TTAA
B.	Restriction enzyme	Ligase	TTAA	AATT	TTAA
C.	Ligase	Restriction enzyme	AATT	TTAA	AATT
D.	Restriction enzyme	Ligase	AATT	TTAA	AATT

30. The data below shows the results of electrophoresis of PCR fragments amplified using primers for the site which has been shown to be altered in Huntington's disease. The inherited mutation in the Huntington's disease gene abnormally repeats the nucleotide sequence CAG from 36 to more than 120 times.

The male parent, as shown by the black box, got Huntington's disease when he was 40 years old. Six of his children (individuals 3, 5, 7, 8, 10, 11) suffer from Huntington's disease, and the age at which the symptoms first began is shown by the number below the band from the PCR fragment.



What is the likely outcome for the normal individuals **4**, **6**, and **9**?

A. Individuals 4 and 9 do not have the trait, and will not get Huntington's disease, but individual 6 is likely to start the disease when he reaches his father's age of 40.

B. Individuals 4, 6, and 9 have not inherited the defect causing Huntington's disease.

C. Individuals 4, 6, and 9 will still develop Huntington's disease at some point in their lives, since the disease is inherited as a dominant trait.

D. Two of the three will develop the disease, since it is inherited as a dominant trait, but the data does not allow you to predict which two.

***** End of Paper 1 *****